Difference Between SPA and MPA and Choosing the Best for our chosen Project

Single Page Application (SPA)

A Single Page Application is a web application that loads a single HTML page and dynamically updates its content without reloading the entire page when the user navigates between different sections.

Advantages:

- Provides a fast and smooth user experience after the initial loading
- Updates parts of the page without reloading the entire page
- Usually communicates with the backend through APIs.

Examples: Gmail

Multi-Page Application (MPA)

A Multi-Page Application is a web application that consists of multiple pages. When the user navigates to another page, the browser usually sends a request to the server and loads a new HTML page.

Advantages:

- Suitable for websites with many independent pages
- Works naturally with server-side rendering
- Each page can have its own URL and content
- Can be developed using technologies such as ASP.NET Core MVC

Examples: Traditional e-commerce websites and content-based websites.

Choosing the Best Approach for Each Project

Project 1: San3ty — Home Maintenance Services Platform

Recommended approach: MPA using ASP.NET Core MVC.

San3ty allows customers to browse maintenance services, search for technicians, make bookings, and track booking statuses. It also includes separate dashboards for customers, technicians, and administrators.

Why MPA is suitable:

- ASP.NET Core MVC supports server-rendered pages through Controllers and Views.

- The project contains many separate pages, such as service listings, technician details, booking pages, and dashboards.

- It integrates naturally with the .NET backend and SQL Server database.

- It is a practical choice for a team working with ASP.NET Core MVC.

+JavaScript and AJAX can still be used to update specific parts of a page without reloading the entire page.